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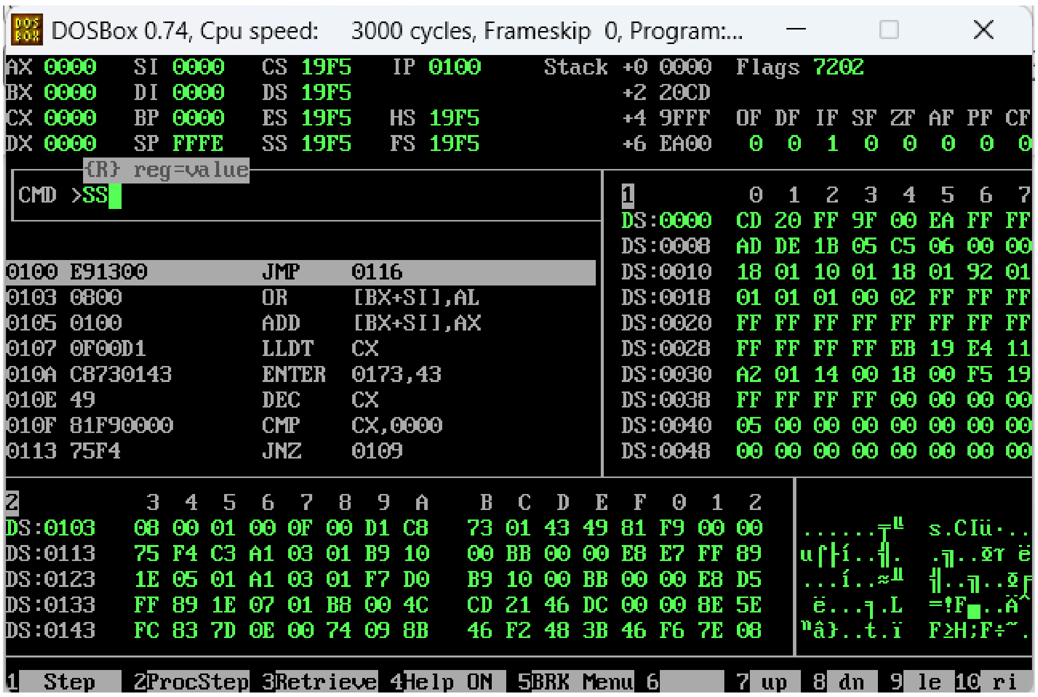
22F-3277
BCS-3C
LAB 6

Q1: Write a program that Count the number of one bits in DX and its complement and check either the both count equal or not.

```
org 0x100

jmp start
        num: dw 8               ;0103 (the address of labels)
        count1: dw 0            ;0105,,,to store original number 1's count
        count2: dw 0            ;0107,,,to store compliment number 1's count
```


```
count:
        ror ax,1
        jnc skip
        inc bx

skip:
        dec cx
        cmp cx,0
        jne count
        ret

start:
        mov ax, [num]
        mov cx, 16
        mov bx, 0
        call count
        mov [count1], bx

        mov ax, [num]
        not ax
        mov cx, 16
        mov bx, 0
        call count
        mov [count2], bx

        mov ax,0x4c00
        int 0x21
```



Q2. Write a Subroutine that will allocate grades according to the marks entered as below. (Setup a memory variable for marks and grade)

Marks Grade

90-100 AA

80-89 AA

70-79 BB

60-69 CC

50-59 DD

40-49 EE

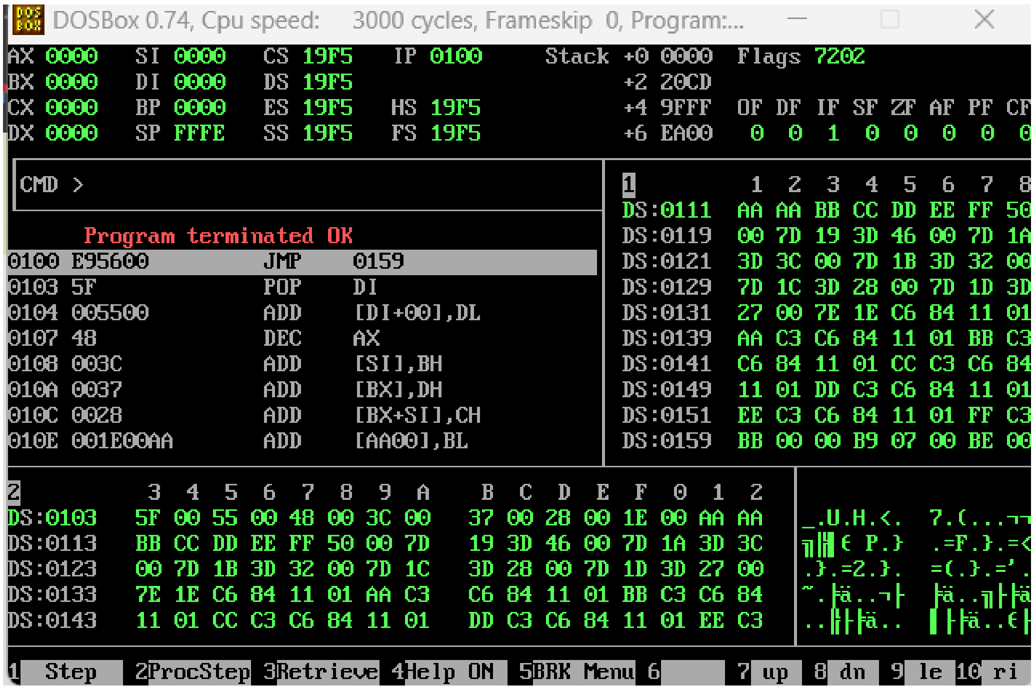
<=39 FF

```
org 0x100

jmp start
        marks: dw 95,85,72,60,55,40,30
        grade: db 0,0,0,0,0,0

grading:
        cmp ax, 80
        jge aa
        cmp ax, 70              ;line 10
        jge bb
        cmp ax, 60
        jge cc
        cmp ax, 50
        jge ddg                 ;line 15
        cmp ax, 40
        jge ee
        cmp ax, 39
        jle ff

aa:
        mov byte [grade+si],0xAA
        ret
bb:
        mov byte [grade+si],0xBB
        ret
cc:
        mov byte [grade+si],0xCC
        ret
ddg:                                    ;line 30
        mov byte [grade+si],0xDD                ;line 31
        ret
ee:
        mov byte [grade+si],0xEE
        ret
ff:
        mov byte [grade+si],0xFF
        ret

start:                  ;line 40

mov bx, 0
mov cx, 7
```

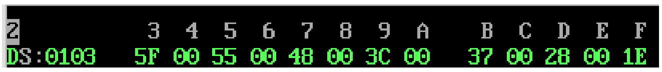
mov si, 0

l1: ;line 46

```
mov ax, [marks+bx]
call grading
inc si
add bx,2
dec cx
cmp cx, 0
jnz l1     ;line 54

mov ax,0x4c00
int 0x21
```



Marks label starts from **0103** to **010F** in m2



Grade label starts from **0111** to **0117** in m1 with grade of every mark in corresponding index